Space Weather Evidence Mapper — PRD

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Overview

We will build a classical-NLP pipeline that reads space-science abstracts and extracts **which space-weather event affects which technological system, and how**. Every claim it extracts links back to the sentence that supports it. The output is an evidence matrix of event × system. Each cell shows how many papers report an effect, whether they agree, and the supporting sentences.

Problem. Findings on how geomagnetic storms and solar particle events affect GNSS, HF radio, satellites and power grids are spread across thousands of papers. Different communities use different terms for the same effect ("scintillation" vs "signal fading", "SEU" vs "single-event upset"). Today a researcher has to read the papers by hand to see where findings agree, conflict or are missing.

What this is not. It does not forecast space weather. It does not judge whether a paper is correct. It does not use an LLM as the core extractor.

Course context. Mini project for Natural Language Processing (702AI0C009), Sem VII, 2026–27. It is worth 10 marks: 5 for the midterm review (topic, scope, objectives) and 5 for the final review (implementation and presentation). The team is 2–3 students. The design deliberately uses syllabus methods from Units 2–7 (CO2–CO4).

Goals and success criteria

The project succeeds if Phase 1 runs end to end on a held-out test set and reports honest, measured numbers. Phases 2–3 are a bonus, not a requirement.

Goals

1. Extract (event, system, effect) triples from abstracts using syllabus methods only.
2. Group different terms for the same thing (synonyms and abbreviations) so the evidence for each pair can be combined.
3. Retrieve the supporting sentences for any event–system query.
4. Produce an evidence matrix that shows agreement, conflict and gaps.
5. Evaluate against a hand-labelled gold set and report where the system fails.

Non-goals

- Forecasting, physical modelling, or judging how good a paper is.
- Parsing full-text PDFs (Phase 3 at most).
- A production web app. A notebook plus a simple Streamlit view is enough.
- Fine-tuning large models as the main approach.

Success criteria (Phase 1)

Criterion	Target
Corpus collected and filtered	⩾ 500 relevant abstracts
Annotator agreement on the gold set	Cohen's κ ⩾ 0.6 for event, system and effect
Entity extraction (event, system)	Precision ⩾ 0.75; recall reported, no target
Triple extraction	F1 reported against the gold set, beating a co-occurrence baseline
Evidence retrieval	Precision@5 reported on ⩾ 20 queries
Demo	Query → matrix cell → supporting sentences, live

The targets are a minimum for an honest result, not a promise of quality. If a number comes in below its target, we report it and analyse why.

Scope and phasing

Phase 1 is the committed scope and the minimum we demo at the final review. Phases 2 and 3 are optional extensions, attempted only if Phase 1 passes its gate with time left.

Phase	Status	Adds	Gate to move on
1. Core evidence mapper	Committed	Corpus, entity tagging (event, system), effect extraction, synonym grouping, retrieval, evidence matrix, evaluation	Pipeline runs end to end, κ ⩾ 0.6, numbers reported

Phase	Status	Adds	Gate to move on
2. Context slots	Optional	**Condition** spans (phase, orbit, Kp threshold) and **uncertainty** (hedge detection)	$\kappa \geq 0.5$ on each new slot, otherwise drop that slot
3. Full text and learned models	Optional	arXiv LaTeX/HTML full text, labels generated by rules (weak supervision), CRF and logistic-regression extractors, coreference	Phase 2 done with ≥ 3 weeks before the final

Rule for the reviews. At the midterm, present all three phases, with Phase 1 explicitly marked as committed. At the final, demo Phase 1 at minimum, plus whatever later phases passed their gates.

Hard stop. Two weeks before the final review we stop adding features. That time goes to evaluation, error analysis, the report and the demo.

Data

The Phase 1 corpus is 500–1,000 English abstracts from the NASA ADS API, restricted to two event families and four system families.

Sources

Source	Used for	Access	Phase
NASA ADS API	Abstracts, titles, year, DOI, bibcode	Free account + API token; rate-limited	1
arXiv `physics.space-ph` (LaTeX or ar5iv HTML)	Full text	Public	3
NASA Technical Reports Server (NTRS)	Operational reports	Public API; PDFs need GROBID	3, optional

Domain restriction

- **Events:** geomagnetic storm (including Kp/Dst levels, CME-driven storms) and solar energetic particle (SEP) events, including solar flares as the triggering cause.

- **Systems:** GNSS/GPS, HF radio, satellites (surface charging, single-event effects, atmospheric drag) and power grids (GICs, transformers).
- **Effects:** degradation, outage/failure, no significant effect, mitigated.

Corpus build

1. Run ADS queries that combine event terms and system terms, for 2000–2026.
2. Remove duplicates by bibcode and DOI, and drop abstracts that aren't in English or are under 50 words.
3. Score relevance with TF-IDF cosine similarity against a seed set of 30 hand-picked abstracts. Hand-check a sample of 50 near the cut-off.
4. Freeze the corpus as versioned JSONL with the query strings logged, so the dataset can be rebuilt.

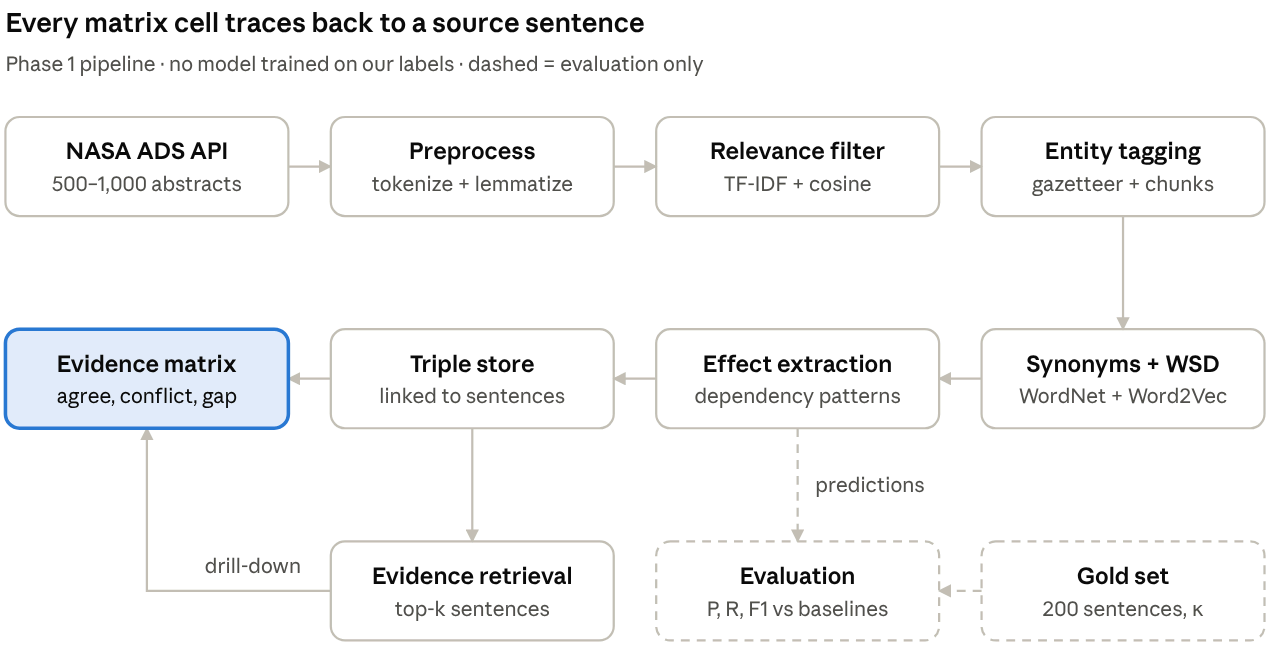
Schema (one record per extracted triple)

```
{"bibcode": "…", "sentence_id": 3, "sentence": "…", "event": {"text": "…",
"canonical": "geomagnetic_storm"}, "system": {"text": "…", "canonical":
"gnss"}, "effect": "degradation", "method": "dep_pattern_07", "confidence":
0.8}
```

Every triple keeps its bibcode and sentence ID, so any cell in the matrix can be traced back to the exact sentence.

System design (Phase 1)

Phase 1 is a fixed, rule-and-embedding pipeline. Every stage is a separate, testable step, and Phase 1 trains no model on our own labels: the hand-labelled gold set is used only to measure performance.



Phase 1 pipeline · 9 stages + evaluation

Abstracts flow left to right, then back through grouping and extraction into a store where every triple keeps its source sentence. The gold set only ever touches evaluation.

#	Stage	Method	Syllabus unit	Output
1	Preprocessing	Sentence and word tokenization, lowercasing, stopword removal, lemmatization (NLTK/spaCy); regex for Kp/Dst values and abbreviations	Unit 2, Unit 3 (regex)	Clean sentences
2	Relevance filter	BoW and TF-IDF vectors, cosine and Jaccard similarity against the seed set	Unit 2	Filtered corpus
3	Domain language model	Bigram/trigram model with Laplace and Kneser-Ney smoothing; compare perplexity on space-weather text vs general astrophysics text	Unit 3	Shows how different the domain's language is

#	Stage	Method	Syllabus unit	Output
4	Entity tagging	Gazetteer + POS-tag and chunk patterns for EVENT and SYSTEM; compare against spaCy's default NER as a baseline	Unit 4	Tagged spans
5	Synonym grouping	WordNet synonym/hypernym lookup + Word2Vec trained on the corpus (with GloVe for comparison); cluster terms into canonical labels	Unit 5	Canonical entity IDs
6	Sense disambiguation	Lesk-style WSD for ambiguous terms ("storm", "charging", "drag", "event")	Unit 5	Ambiguous terms resolved to the right meaning
7	Effect extraction	Dependency-parse patterns linking EVENT → verb → SYSTEM ("X caused/degraded/disrupted Y"); an effect-verb word list maps to the four effect labels; negation handling	Unit 4	Triples
8	Evidence store and retrieval	TF-IDF index over sentences; a query (event, system) returns the top-k supporting sentences	Unit 7 (IR)	Ranked evidence
9	Evidence matrix	Combine triples per (event, system) pair: paper count, split by effect label, agreement score	Unit 7 (IE)	Matrix + drill-down

Stack. Python 3.11, NLTK, spaCy (`en_core_web_sm`), gensim, scikit-learn, pandas and Streamlit, plus doccano or Label Studio for annotation. Everything runs on a laptop CPU. Fix the random seeds and pin package versions so results can be reproduced.

Evaluation plan

All numbers come from a gold set we freeze before tuning any rules, so we never tune against the test set.

Gold set

- About 300 sentences from about 150 abstracts, stratified across the 2 event families × 4 system families.
- Split: 100 sentences for development (we tune rules on these) and 200 for test (touched once, at the end).
- Two annotators label every test sentence independently, following a 1–2 page guideline with examples for each label. Disagreements are resolved by discussion after κ is computed.
- Report Cohen's κ for each slot. If κ is below 0.6, revise the guideline, re-annotate a fresh batch, and report both rounds.

Metrics and baselines

Task	Metric	Baseline it must beat
Entity tagging	Span-level precision, recall, F1 (exact and partial match)	spaCy's default NER; gazetteer lookup with no POS/chunk rules
Synonym grouping	Purity of term clusters on 50 hand-checked term pairs	WordNet only
Effect extraction	Triple-level precision, recall, F1	Sentence co-occurrence (event and system in the same sentence = a relation)
Retrieval	Precision@5, Recall@10 on 20–25 written queries	BoW without TF-IDF weighting
Language model	Perplexity, bigram vs trigram, with and without smoothing	Unsmoothed bigram

Ablations. Remove one stage at a time (WSD, synonym grouping, negation handling) and report how F1 changes. This shows each syllabus technique earns its place.

Error analysis. Hand-sort 50 false positives and 50 false negatives into types: missed synonym, wrong parse, hedged claim, an effect spread across sentences, negation. This feeds directly into the case for Phase 2 and 3.

Optional LLM comparison. Run a prompted LLM on the same test sentences, *only* with faculty approval. Report it as a comparison point, never as the system.

Extensions (Phases 2 and 3)

Each extension plugs into the Phase 1 pipeline without changing what it already outputs, so dropping one never breaks the demo.

Phase 2: context slots

- **Uncertainty:** detect hedge cues using a word list ("may", "suggests", "is likely", "could not be confirmed") plus a logistic-regression classifier on n-gram and POS features. Each triple gets a label: asserted, hedged or negated. The matrix can then separate strong evidence from tentative evidence.
- **Condition:** extract spans from chunk patterns and parse subtrees for storm phase ("main phase", "recovery phase"), location or orbit ("LEO", "GEO", "high latitudes") and thresholds ("Kp ⩾ 7", "Dst < −100 nT"). Evaluate as span F1.
- Both slots get added to the gold-set guideline, with κ reported for each. If either slot's κ is below 0.5, drop that slot and report why.

Phase 3: full text and learned models

- **Full text:** take LaTeX source or ar5iv HTML from arXiv `physics.space-ph`, and extract only the Results and Discussion sections. Run NTRS PDFs through GROBID only if needed. Don't write a custom PDF parser.
- **Weak supervision:** 15–25 rules that each label sentences automatically, combined by majority vote (or Snorkel), to produce training labels at scale. The gold set stays test-only.
- **Learned extractors:** a CRF on POS, chunk and gazetteer features for entity tagging; logistic regression on dependency-path and Word2Vec features for relation classification. Compare against the Phase 1 rules on the same test set.
- **Coreference:** link "this storm", "it" and "the event" back to their antecedent, so effects reported in a later sentence still attach to the right event. Report the gain in recall.

Timeline and review deliverables

The plan is written in relative weeks from kickoff, because the review dates aren't confirmed yet. Phase 1 takes 6 weeks. Anything beyond that goes to Phase 2.

1. **Week 1:** get the ADS token, write and log the queries, pull and deduplicate the corpus, pick the seed set, draft the annotation guideline.
2. **Week 2:** preprocessing, TF-IDF relevance filter, n-gram language model and perplexity. Annotate the 100 development sentences. **Gate:** ⩾ 500 relevant abstracts, or narrow the query.
3. **Week 3:** gazetteer and POS/chunk entity tagger, Word2Vec plus WordNet synonym grouping, WSD. Both annotators label the 200 test sentences; compute κ. **Gate:** κ ⩾ 0.6.
4. **Week 4:** dependency-pattern effect extraction, negation handling, triple store.
5. **Week 5:** retrieval index, evidence matrix, Streamlit view. **Gate:** pipeline runs end to end.
6. **Week 6:** run on the test set once, baselines, ablations, error analysis, report and slides.

What each review needs

Review	Marks	We bring
Midterm (topic, scope, objectives)	5	Problem statement, this PRD's scope and phasing, the domain restriction, the sources confirmed working, a sample of 20 abstracts with hand-marked triples, and the evaluation plan
Final (implementation, presentation)	5	Live demo (query → matrix → evidence), results tables against baselines, κ, ablations, error analysis, the limitations slide, and any Phase 2–3 work that passed its gate

Who does what (for a team of 3; merge roles if there are 2)

- **Data and annotation lead:** corpus, guideline, gold set, κ.
- **Extraction lead:** entity tagging, synonym grouping, WSD, effect patterns.
- **Evaluation and demo lead:** retrieval, matrix, baselines, ablations, Streamlit.

Everyone annotates, so we have two independent annotators for every test sentence.

Risks, policy and open questions

The two biggest risks are inconsistent annotation and low recall from rules. Both are handled by measuring early and reporting honestly, not by adding complexity.

Risk	Likelihood	Impact	Mitigation
Annotators disagree (κ < 0.6)	Medium	Gold set can't be trusted	Guideline with examples, a pilot batch of 30, revise and re-annotate; narrow the label set if needed
Rules give high precision but low recall	High	Low F1	Expected; report the precision/recall trade-off; the error analysis justifies Phase 3 learned models
ADS API limits or query drift	Low	Corpus delays	Cache all responses, log the queries, freeze the corpus in week 2
Abstracts rarely state effects explicitly	Medium	Sparse matrix	Treat the gaps as a finding; widen to full text only in Phase 3
Scope creep into Phases 2–3	Medium	Phase 1 unfinished	Gates and the hard stop are fixed now and not renegotiated
Domain knowledge gaps	Medium	Wrong labels	Use NOAA's space weather scales and a review paper as the label reference; ask a physics faculty member to check 20 labels

Academic integrity and GenAI. The course policy allows GenAI for background research only, and requires original work. So:

- All pipeline code is written by the team. Any AI-assisted background reading is noted in the report.
- No LLM produces the gold labels or the extracted triples.
- An LLM comparison runs only with written approval from the course faculty.

Open questions

- [ ] Confirm the midterm and final review dates with the faculty.
- [ ] Confirm team size (2 or 3) and assign the roles.
- [ ] Ask the faculty whether an LLM comparison is allowed.

- [ ] Confirm the event and system families with one domain reference before writing the gazetteer.